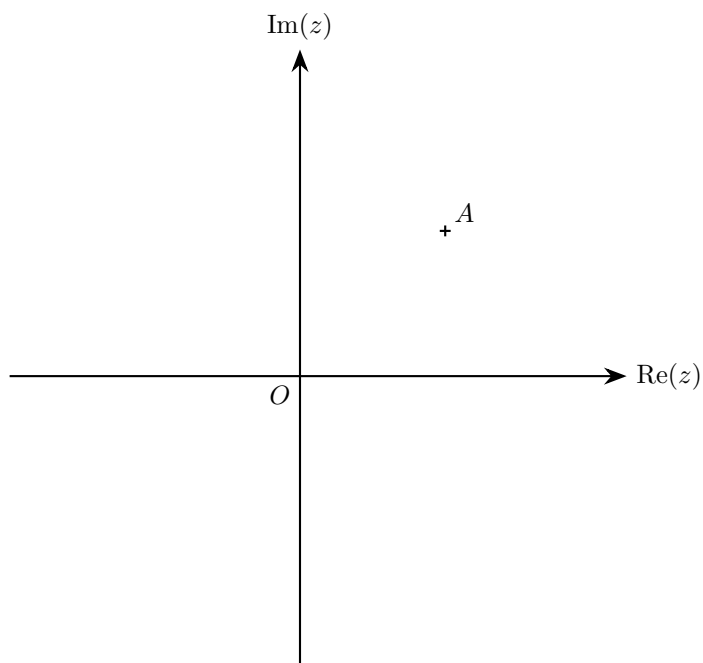


## Questions

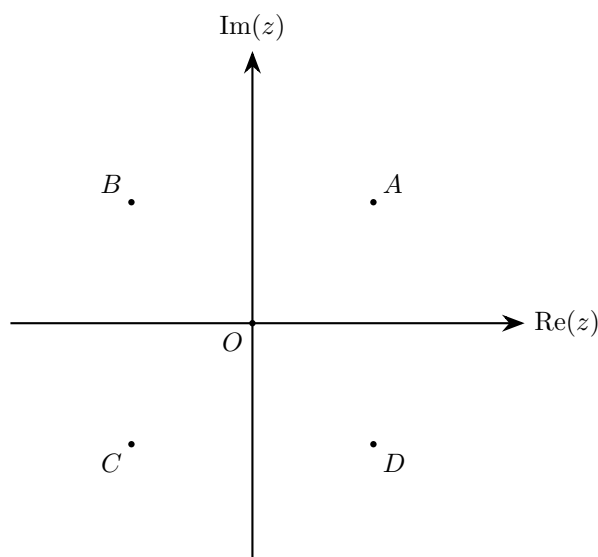
|             |          |    |
|-------------|----------|----|
| Question 1  | 8 marks  | 2  |
| Question 2  | 10 marks | 4  |
| Question 3  | 13 marks | 6  |
| Question 4  | 10 marks | 9  |
| Question 5  | 5 marks  | 11 |
| Question 6  | 10 marks | 13 |
| Question 7  | 7 marks  | 15 |
| Question 8  | 12 marks | 17 |
| Question 9  | 9 marks  | 20 |
| Question 10 | 9 marks  | 23 |
| Question 11 | 14 marks | 25 |
| Question 12 | 8 marks  | 28 |
| Question 13 | 11 marks | 30 |



1. An Argand diagram with the point  $A$  representing a complex number  $z_1$  is shown in the diagram. The complex numbers  $z_2$ ,  $z_3$  and  $z_4$  are  $z_1 e^{\frac{1}{2}\pi i}$ ,  $z_1 e^{\pi i}$  and  $z_1 e^{\frac{3}{2}\pi i}$  respectively.
- (a) (i) On a copy of the Argand diagram, mark the points  $B$ ,  $C$  and  $D$  representing the complex numbers  $z_2$ ,  $z_3$  and  $z_4$ . [2]
- (ii) Show that  $z_1 + z_2 + z_3 + z_4 = 0$ . [2]
- (b) Given now that  $z_1$ ,  $z_2$ ,  $z_3$  and  $z_4$  are roots of the equation  $z^4 = -16$ , find these four roots, giving your answers in the form  $a + ib$ , where  $a$  and  $b$  are real and exact. [4]

### Solution

- (a) (i) Multiplying  $z_1$  by  $e^{i\theta}$  corresponds to rotating  $z_1$  by  $\theta$  anticlockwise about the origin. So  $B$ ,  $C$  and  $D$  are the images of  $A$  under such rotations of  $90^\circ$ ,  $180^\circ$  and  $270^\circ$  respectively.



(ii)

$$\begin{aligned} z_1 + z_2 + z_3 + z_4 &= z_1 + z_1 e^{\frac{1}{2}\pi i} + z_1 e^{\pi i} + z_1 e^{\frac{3}{2}\pi i} \\ &= z_1 + iz_1 - z_1 - iz_1 \\ &= z_1(1 + i - 1 - i) \\ &= 0 \end{aligned}$$

Hence  $z_1 + z_2 + z_3 + z_4 = 0$ .

(Alternatively, use that  $1 + e^{\frac{1}{2}\pi i} + e^{\pi i} + e^{\frac{3}{2}\pi i}$  is a geometric sum with common ratio  $r = e^{\frac{1}{2}\pi i}$ )

(b) Write

$$\begin{aligned} -16 &= 16e^{i\pi} \\ &= 16e^{i(\pi+2k\pi)} \end{aligned}$$

so the fourth roots are

$$z = 2e^{i\left(\frac{\pi+2k\pi}{4}\right)} \quad k = 0, 1, 2, 3$$

Hence

$$\begin{aligned} k = 0 : \quad z &= 2e^{i\pi/4} = 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = \sqrt{2} + i\sqrt{2} \\ k = 1 : \quad z &= 2e^{3i\pi/4} = 2\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} + i\sqrt{2} \\ k = 2 : \quad z &= 2e^{5i\pi/4} = 2\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} - i\sqrt{2} \\ k = 3 : \quad z &= 2e^{7i\pi/4} = 2\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \sqrt{2} - i\sqrt{2} \end{aligned}$$

Therefore the four roots are

$$\sqrt{2} + i\sqrt{2}, \quad -\sqrt{2} + i\sqrt{2}, \quad -\sqrt{2} - i\sqrt{2}, \quad \sqrt{2} - i\sqrt{2}$$

2. The five solutions of the equation  $z^5 = 16 + 16\sqrt{3}i$  are represented by points on an Argand diagram.

These points are the vertices of a regular pentagon  $P$  with centre at the origin.

- (a) Determine the five complex numbers that represent these vertices, giving your answers in the form  $re^{i\theta}$ , where  $0 \leq \theta < 2\pi$ . [4]
- (b) Sketch  $P$  on an Argand Diagram, taking care to label each vertex. [2]
- (c) Show that the area of  $P$  can be written  $k \sin \frac{2\pi}{5}$  where  $k$  is an integer to be determined. [4]

### Solution

(a) Let

$$w = 16 + 16\sqrt{3}i$$

First write  $w$  in polar form.

Its modulus is

$$\begin{aligned} |w| &= \sqrt{16^2 + (16\sqrt{3})^2} \\ &= \sqrt{256 + 768} \\ &= \sqrt{1024} \\ &= 32 \end{aligned}$$

Its argument is

$$\arg(w) = \tan^{-1}\left(\frac{16\sqrt{3}}{16}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

since  $w$  lies in the first quadrant.

So

$$16 + 16\sqrt{3}i = 32e^{i\pi/3}$$

Now solve

$$z^5 = 32e^{i\pi/3}$$

The fifth roots have modulus

$$32^{1/5} = 2$$

and arguments

$$\theta = \frac{\frac{\pi}{3} + 2\pi k}{5} = \frac{\pi}{15} + \frac{2\pi k}{5} \quad (k = 0, 1, 2, 3, 4)$$

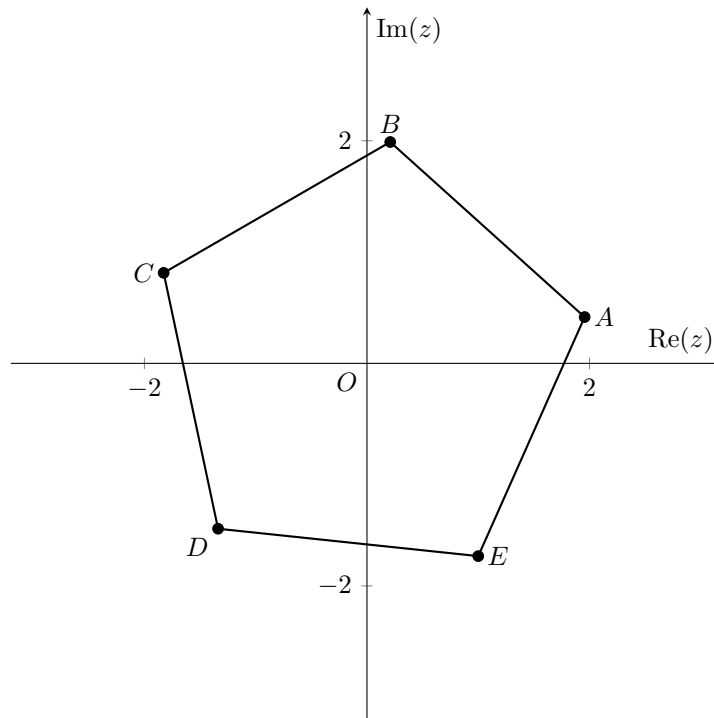
Hence the five roots are

$$\begin{aligned} z &= 2e^{i\pi/15} \\ z &= 2e^{i7\pi/15} \\ z &= 2e^{i13\pi/15} \\ z &= 2e^{i19\pi/15} \\ z &= 2e^{i25\pi/15} = 2e^{i5\pi/3} \end{aligned}$$

(b) Let the vertices in anticlockwise order be

$$A = 2e^{i\pi/15}, \quad B = 2e^{i7\pi/15}, \quad C = 2e^{i13\pi/15}, \quad D = 2e^{i19\pi/15}, \quad E = 2e^{i5\pi/3}$$

They all lie on the circle of radius 2 centred at the origin, with equal angular spacing  $\frac{2\pi}{5}$ .



- (c) Join the origin to each vertex. This splits the pentagon into 5 congruent isosceles triangles. Each vertex has modulus 2, so each triangle has two sides of length 2. Also, the arguments of adjacent vertices differ by

$$\frac{2\pi}{5}$$

so the angle at the origin in each triangle is  $\frac{2\pi}{5}$ .

Hence the area of one triangle is

$$\frac{1}{2} \times 2 \times 2 \times \sin \frac{2\pi}{5} = 2 \sin \frac{2\pi}{5}$$

Therefore the area of the pentagon  $P$  is

$$5 \left( 2 \sin \frac{2\pi}{5} \right) = 10 \sin \frac{2\pi}{5}$$

So it can be written in the form  $k \sin \frac{2\pi}{5}$  with

$$k = 10$$

3. (i) Find the 8th roots of  $-128 + 128\sqrt{3}i$  in the form  $re^{i\theta}$ , where  $r > 0$  and  $0 \leq \theta < 2\pi$ . [5]
- (ii) These 8th roots are represented in the Argand diagram, in order of increasing  $\theta$ , by the points  $A, B, C, D, E, F, G, H$ .  
Draw the Argand diagram, making clear which point is which. [2]
- (iii) The mid-point of  $BD$  is the point  $P$  which represents the complex number  $w$ .  
Find, in exact form, the modulus and argument of  $w$ . [3]
- (iv)  $w$  is an  $n$ th root of a real number  $a$ , where  $n$  is a positive integer. State the least possible value of  $n$  and find the corresponding value of  $a$ . [3]

### Solution

(i) Let

$$z = -128 + 128\sqrt{3}i$$

Its modulus is

$$\begin{aligned} |z| &= \sqrt{(-128)^2 + (128\sqrt{3})^2} \\ &= 128\sqrt{1+3} \\ &= 256 \end{aligned}$$

Also,

$$\tan \theta = \frac{128\sqrt{3}}{-128} = -\sqrt{3}$$

Since  $z$  lies in the second quadrant, its argument is

$$\theta = \frac{2\pi}{3}$$

So

$$-128 + 128\sqrt{3}i = 256e^{i2\pi/3}$$

If  $z_k$  is an 8th root, then

$$z_k^8 = 256e^{i2\pi/3}$$

Hence the modulus of each root is

$$256^{1/8} = 2$$

and the arguments are

$$\frac{\frac{2\pi}{3} + 2k\pi}{8} = \frac{\pi}{12} + \frac{k\pi}{4}, \quad k = 0, 1, \dots, 7$$

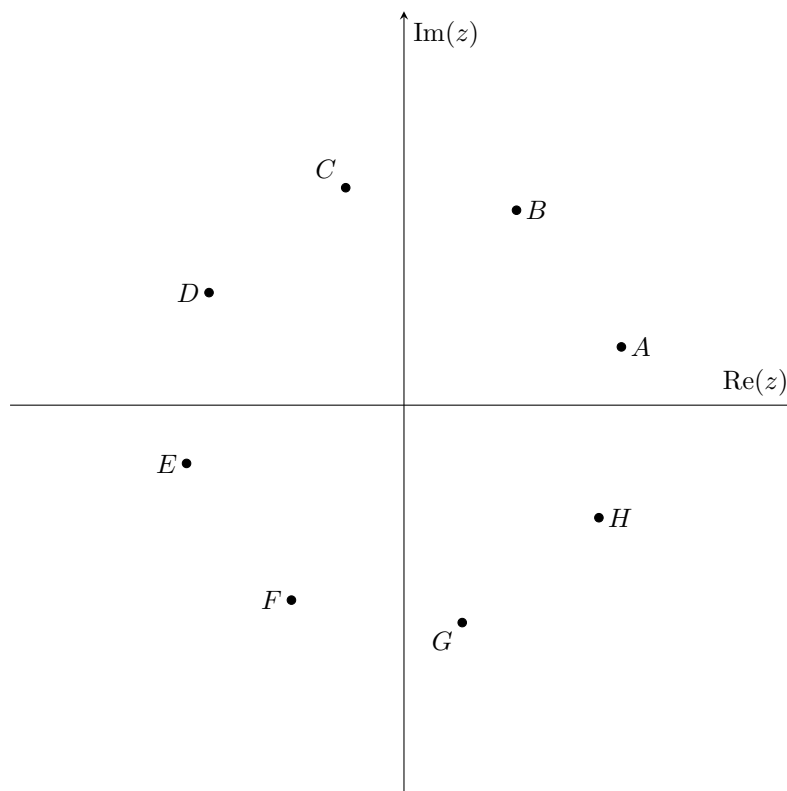
Therefore the 8th roots are

$$z_k = 2e^{i(\pi/12 + k\pi/4)}, \quad k = 0, 1, \dots, 7$$

(ii) In increasing order of argument, the points are

$$A : \frac{\pi}{12}, \quad B : \frac{\pi}{3}, \quad C : \frac{7\pi}{12}, \quad D : \frac{5\pi}{6}, \quad E : \frac{13\pi}{12}, \quad F : \frac{4\pi}{3}, \quad G : \frac{19\pi}{12}, \quad H : \frac{11\pi}{6}$$

All eight points lie on the circle  $|z| = 2$ .



(iii) The complex numbers represented by  $B$  and  $D$  are

$$2e^{i\pi/3} \quad \text{and} \quad 2e^{i5\pi/6}$$

Since  $P$  is the midpoint of  $BD$ , the complex number  $w$  is

$$\begin{aligned} w &= \frac{2e^{i\pi/3} + 2e^{i5\pi/6}}{2} \\ &= e^{i\pi/3} + e^{i5\pi/6} \\ &= e^{i7\pi/12} \left( e^{-i\pi/4} + e^{i\pi/4} \right) \\ &= e^{i7\pi/12} \left( 2 \cos \frac{\pi}{4} \right) \\ &= \sqrt{2} e^{i7\pi/12} \end{aligned}$$

So the modulus and argument of  $w$  are

$$|w| = \sqrt{2}, \quad \arg(w) = \frac{7\pi}{12}$$

(iv) Since

$$w = \sqrt{2} e^{i7\pi/12}$$

we have

$$w^n = (\sqrt{2})^n e^{i7n\pi/12}$$

For  $w^n$  to be real, its argument must be a multiple of  $\pi$ , so

$$\frac{7n\pi}{12} = m\pi$$

for some integer  $m$ . Hence

$$\frac{7n}{12} \in \mathbb{Z}$$

As 7 and 12 have no common factor, the least possible value is

$$n = 12$$

Then

$$\begin{aligned}a &= w^{12} \\&= (\sqrt{2})^{12} e^{i7\pi} \\&= 2^6(-1) \\&= -64\end{aligned}$$

So the least possible value of  $n$  is 12, and the corresponding value of  $a$  is  $-64$ .

4. Given that  $z_1 = 7 - i$  and  $z_2 = \frac{z_1}{1 - i}$ ,

(a) find  $z_2$  in the form  $a + bi$ , where  $a$  and  $b$  are real. [2]

(b) Show on an Argand diagram the point  $P$  representing  $z_1$  and the point  $Q$  representing  $z_2$ . [2]

(c) Given that  $O$  is the origin, show that  $\angle OQP = \frac{\pi}{2}$ . [2]

The circle passing through the points  $O$ ,  $P$  and  $Q$  has centre  $C$ . Find

(d) the complex number represented by  $C$ . [2]

(e) the exact value of the radius of the circle. [2]

### Solution

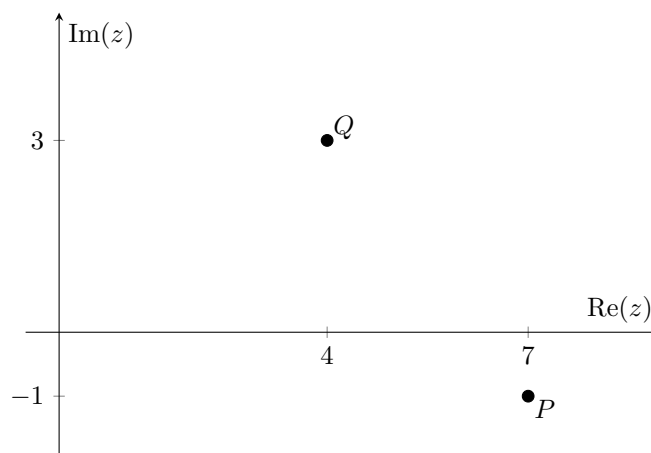
(a) Rationalise the denominator:

$$\begin{aligned} z_2 &= \frac{7 - i}{1 - i} \cdot \frac{1 + i}{1 + i} \\ &= \frac{(7 - i)(1 + i)}{(1 - i)(1 + i)} \\ &= \frac{7 + 7i - i - i^2}{1 - i^2} \\ &= \frac{7 + 6i + 1}{1 + 1} \\ &= \frac{8 + 6i}{2} \\ &= 4 + 3i \end{aligned}$$

(b) On an Argand diagram, the complex number  $x + yi$  is represented by the point  $(x, y)$ .

So

$$z_1 = 7 - i \Rightarrow P(7, -1), \quad z_2 = 4 + 3i \Rightarrow Q(4, 3)$$



(c) Using the coordinates from part (b),

$$O = (0, 0), \quad P = (7, -1), \quad Q = (4, 3)$$

Take the vectors from  $Q$ :

$$\overrightarrow{QO} = \begin{pmatrix} -4 \\ -3 \end{pmatrix}, \quad \overrightarrow{QP} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

Their scalar product is

$$\begin{aligned}\overrightarrow{QO} \cdot \overrightarrow{QP} &= (-4)(3) + (-3)(-4) \\ &= -12 + 12 \\ &= 0\end{aligned}$$

So  $\overrightarrow{QO}$  is perpendicular to  $\overrightarrow{QP}$ , which means

$$\angle OQP = \frac{\pi}{2}$$

(d) Since  $\angle OQP = \frac{\pi}{2}$ , the side  $OP$  is a diameter of the circle through  $O$ ,  $P$  and  $Q$ .

Therefore the centre  $C$  is the midpoint of  $O(0, 0)$  and  $P(7, -1)$ :

$$\begin{aligned}C &= \left( \frac{0+7}{2}, \frac{0+(-1)}{2} \right) \\ &= \left( \frac{7}{2}, -\frac{1}{2} \right)\end{aligned}$$

So the complex number represented by  $C$  is

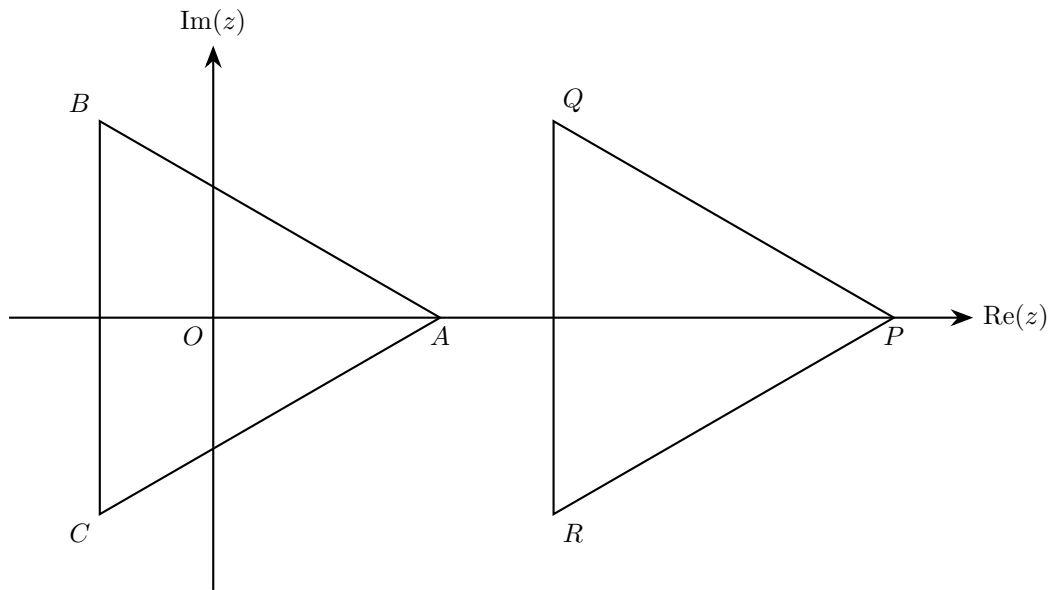
$$\frac{7}{2} - \frac{1}{2}i$$

(e) The diameter is  $OP$ , so first find its length:

$$\begin{aligned}OP &= \sqrt{(7-0)^2 + (-1-0)^2} \\ &= \sqrt{49+1} \\ &= \sqrt{50} \\ &= 5\sqrt{2}\end{aligned}$$

Hence the radius of the circle is

$$\frac{1}{2}OP = \frac{5\sqrt{2}}{2}$$



5. The cube roots of unity are represented on the Argand diagram above by the points  $A$ ,  $B$  and  $C$ .

The complex numbers represented by  $P$ ,  $Q$  and  $R$  are obtained by adding 2 to the complex numbers represented by  $A$ ,  $B$  and  $C$  respectively.

Determine a degree 6 polynomial equation with integer coefficients whose roots are the complex numbers represented by the points  $A$ ,  $B$ ,  $C$ ,  $P$ ,  $Q$  and  $R$ .

[5]

### Solution

Let the required polynomial have variable  $z$ .

The points  $A, B, C$  represent the cube roots of unity, so their complex numbers satisfy

$$z^3 = 1$$

Hence  $A, B, C$  are the roots of

$$z^3 - 1 = 0$$

Now  $P, Q, R$  are obtained by adding 2 to the complex numbers at  $A, B, C$ .

So if  $z$  represents one of  $P, Q, R$ , then  $z - 2$  is a cube root of unity. Therefore

$$(z - 2)^3 = 1$$

so

$$(z - 2)^3 - 1 = 0$$

Expanding:

$$\begin{aligned} (z - 2)^3 - 1 &= (z^3 - 6z^2 + 12z - 8) - 1 \\ &= z^3 - 6z^2 + 12z - 9 \end{aligned}$$

So  $P, Q, R$  are the roots of

$$z^3 - 6z^2 + 12z - 9 = 0$$

Therefore a degree 6 polynomial with roots  $A, B, C, P, Q, R$  is

$$(z^3 - 1)(z^3 - 6z^2 + 12z - 9) = 0$$

Now expand:

$$\begin{aligned} (z^3 - 1)(z^3 - 6z^2 + 12z - 9) &= z^3(z^3 - 6z^2 + 12z - 9) - (z^3 - 6z^2 + 12z - 9) \\ &= z^6 - 6z^5 + 12z^4 - 9z^3 - z^3 + 6z^2 - 12z + 9 \\ &= z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9 \end{aligned}$$

Hence the required polynomial equation is

$$z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9 = 0$$

6. In an Argand diagram, the points  $A$ ,  $B$  and  $C$  are the vertices of an equilateral triangle with centre  $G$ . The point  $G$  represents the complex number  $1-2i$ , and the point  $A$  represents the complex number  $5+2i$ .

- (a) Find the complex numbers represented by the points  $B$  and  $C$ , giving your answers in the form  $x+iy$ , where  $x$  and  $y$  are real and exact. [6]

The points  $D$ ,  $E$  and  $F$  are the centroids of triangles  $GBC$ ,  $GCA$  and  $GAB$  respectively.

- (b) Find the exact area of triangle  $DEF$ . [4]

### Solution

- (a) Let

$$z_G = 1 - 2i, \quad z_A = 5 + 2i$$

Then the vector from  $G$  to  $A$  is

$$z_A - z_G = (5 + 2i) - (1 - 2i) = 4 + 4i$$

In an equilateral triangle, the vectors from the centre to the three vertices are  $120^\circ$  apart. So the vectors from  $G$  to  $B$  and  $C$  are obtained by rotating  $4 + 4i$  through  $\pm 120^\circ$ .

Take  $B$  to be the  $+120^\circ$  rotation. Then

$$z_B - z_G = (4 + 4i) \left( -\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

Expanding,

$$\begin{aligned} z_B - z_G &= 4 \left( -\frac{1}{2} \right) + 4 \left( \frac{\sqrt{3}}{2}i \right) + 4i \left( -\frac{1}{2} \right) + 4i \left( \frac{\sqrt{3}}{2}i \right) \\ &= -2 + 2\sqrt{3}i - 2i + 2\sqrt{3}i^2 \\ &= -2 - 2\sqrt{3} + (2\sqrt{3} - 2)i \end{aligned}$$

Hence

$$\begin{aligned} z_B &= z_G + (z_B - z_G) \\ &= (1 - 2i) + [-2 - 2\sqrt{3} + (2\sqrt{3} - 2)i] \\ &= -1 - 2\sqrt{3} + (2\sqrt{3} - 4)i \end{aligned}$$

Now take  $C$  to be the  $-120^\circ$  rotation. Then

$$z_C - z_G = (4 + 4i) \left( -\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

Expanding,

$$\begin{aligned} z_C - z_G &= 4 \left( -\frac{1}{2} \right) + 4 \left( -\frac{\sqrt{3}}{2}i \right) + 4i \left( -\frac{1}{2} \right) + 4i \left( -\frac{\sqrt{3}}{2}i \right) \\ &= -2 - 2\sqrt{3}i - 2i - 2\sqrt{3}i^2 \\ &= -2 + 2\sqrt{3} - (2 + 2\sqrt{3})i \end{aligned}$$

Hence

$$\begin{aligned} z_C &= z_G + (z_C - z_G) \\ &= (1 - 2i) + [-2 + 2\sqrt{3} - (2 + 2\sqrt{3})i] \\ &= -1 + 2\sqrt{3} - (4 + 2\sqrt{3})i \end{aligned}$$

Therefore

$$B = -1 - 2\sqrt{3} + (2\sqrt{3} - 4)i, \quad C = -1 + 2\sqrt{3} - (4 + 2\sqrt{3})i$$

with  $B$  and  $C$  interchangeable.

(b) Since  $G$  is the centre of the equilateral triangle  $ABC$ , it is also the centroid of  $ABC$ . Therefore

$$z_G = \frac{z_A + z_B + z_C}{3}$$

so

$$z_B + z_C = 3z_G - z_A$$

The centroid of triangle  $GBC$  is  $D$ , so

$$\begin{aligned} z_D &= \frac{z_G + z_B + z_C}{3} \\ &= \frac{z_G + (3z_G - z_A)}{3} \\ &= \frac{4z_G - z_A}{3} \end{aligned}$$

Similarly,

$$z_E = \frac{4z_G - z_B}{3}, \quad z_F = \frac{4z_G - z_C}{3}$$

Now compare these with  $z_G$ :

$$\begin{aligned} z_D - z_G &= \frac{4z_G - z_A}{3} - z_G = \frac{z_G - z_A}{3} = -\frac{1}{3}(z_A - z_G) \\ z_E - z_G &= -\frac{1}{3}(z_B - z_G) \\ z_F - z_G &= -\frac{1}{3}(z_C - z_G) \end{aligned}$$

So triangle  $DEF$  is similar to triangle  $ABC$ , with all lengths scaled by a factor of  $\frac{1}{3}$ . Hence the area scale factor is

$$\left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Next, find the area of  $ABC$ .

First,

$$GA = |z_A - z_G| = |4 + 4i| = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

Since  $G$  is the centre of the equilateral triangle, the three angles at  $G$  are equal and sum to  $360^\circ$ , so

$$\angle AGB = 120^\circ$$

Also,

$$GA = GB = GC = 4\sqrt{2}$$

Hence

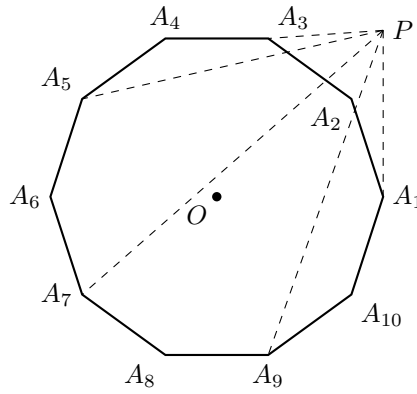
$$\begin{aligned} \text{Area of } \triangle AGB &= \frac{1}{2}(GA)(GB) \sin 120^\circ \\ &= \frac{1}{2}(4\sqrt{2})(4\sqrt{2}) \cdot \frac{\sqrt{3}}{2} \\ &= 16 \cdot \frac{\sqrt{3}}{2} \\ &= 8\sqrt{3} \end{aligned}$$

The triangle  $ABC$  is made up of the three congruent triangles  $AGB$ ,  $BGC$  and  $CGA$ , so

$$\text{Area of } ABC = 3(8\sqrt{3}) = 24\sqrt{3}$$

Therefore

$$\begin{aligned} \text{Area of } DEF &= \frac{1}{9} \times 24\sqrt{3} \\ &= \frac{8\sqrt{3}}{3} \end{aligned}$$



7. (a) Using the identity  $zz^* = |z|^2$ , or otherwise, show that if  $w$  is any root of unity then

$$|w - (1 + i)|^2 = 3 - (1 - i)w - (1 + i)w^* \quad [3]$$

The diagram shows a regular decagon  $A_1A_2 \dots A_{10}$  whose vertices all lie on the unit circle with centre at the origin  $O$  and  $A_1$  at  $(1, 0)$ . The point  $P$  lies in the same plane as the decagon and has coordinates  $(1, 1)$ .

- (b) Using the result from part (a), find

$$\sum_{r=1}^5 (PA_{2r-1})^2 \quad [4]$$

### Solution

- (a) Using

$$|z|^2 = zz^*$$

we have

$$\begin{aligned} |w - (1 + i)|^2 &= (w - (1 + i))(w^* - (1 - i)) \\ &= ww^* - (1 - i)w - (1 + i)w^* + (1 + i)(1 - i) \end{aligned}$$

Now  $w$  is a root of unity, so  $|w| = 1$ . Hence

$$ww^* = |w|^2 = 1$$

Also,

$$(1 + i)(1 - i) = 1 - i^2 = 2$$

So

$$\begin{aligned} |w - (1 + i)|^2 &= 1 - (1 - i)w - (1 + i)w^* + 2 \\ &= 3 - (1 - i)w - (1 + i)w^* \end{aligned}$$

Therefore

$$|w - (1 + i)|^2 = 3 - (1 - i)w - (1 + i)w^*$$

- (b) Let the complex number representing  $A_{2r-1}$  be  $w_r$ .

Since the decagon is regular and  $A_1 = (1, 0)$ , the odd-numbered vertices are at arguments

$$0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

so

$$w_r = e^{2\pi i(r-1)/5} \quad (r = 1, 2, 3, 4, 5)$$

These are exactly the 5th roots of unity.

Also, the point  $P = (1, 1)$  corresponds to the complex number  $1 + i$ . Therefore

$$PA_{2r-1} = |w_r - (1 + i)|$$

and so

$$(PA_{2r-1})^2 = |w_r - (1 + i)|^2$$

Using the result from part (a),

$$(PA_{2r-1})^2 = 3 - (1 - i)w_r - (1 + i)w_r^*$$

Now sum for  $r = 1$  to 5:

$$\begin{aligned}\sum_{r=1}^5 (PA_{2r-1})^2 &= \sum_{r=1}^5 (3 - (1 - i)w_r - (1 + i)w_r^*) \\ &= 15 - (1 - i) \sum_{r=1}^5 w_r - (1 + i) \sum_{r=1}^5 w_r^*\end{aligned}$$

Since  $w_1, w_2, w_3, w_4, w_5$  are the 5th roots of unity,

$$\sum_{r=1}^5 w_r = 0$$

Taking complex conjugates gives

$$\sum_{r=1}^5 w_r^* = 0$$

Hence

$$\sum_{r=1}^5 (PA_{2r-1})^2 = 15$$

8. You are given the complex number  $\alpha = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$  and the equation  $z^5 = -1$ .

(a) Show that  $\alpha$  is a root of the equation. [2]

(b) Write down the other four roots of the equation. [1]

(c) Show that

$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0 \quad [2]$$

(d) Hence show that

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0 \quad [3]$$

(e) Hence determine the value of  $\cos \frac{3\pi}{5}$  in the form  $a + b\sqrt{c}$ , where  $a$ ,  $b$  and  $c$  are rational numbers to be found. [4]

### Solution

(a) We have

$$\alpha = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$$

Using De Moivre's theorem,

$$\begin{aligned} \alpha^5 &= \cos \left(5 \cdot \frac{3\pi}{5}\right) + i \sin \left(5 \cdot \frac{3\pi}{5}\right) \\ &= \cos 3\pi + i \sin 3\pi \\ &= -1 + 0i \\ &= -1 \end{aligned}$$

So  $\alpha$  satisfies  $z^5 = -1$ , and hence  $\alpha$  is a root of the equation.

(b) Since

$$-1 = \cos((2k+1)\pi) + i \sin((2k+1)\pi)$$

the five roots are

$$z = \cos \frac{(2k+1)\pi}{5} + i \sin \frac{(2k+1)\pi}{5} \quad (k = 0, 1, 2, 3, 4)$$

Given that  $\alpha = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$ , the other four roots are

$$\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}, \quad \cos \pi + i \sin \pi, \quad \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}, \quad \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}$$

(c) From part (a),  $\alpha^5 = -1$ , so

$$\alpha^5 + 1 = 0$$

Divide  $x^5 + 1$  by  $x + 1$  to obtain the factorisation

$$x^5 + 1 = (x + 1)(x^4 - x^3 + x^2 - x + 1)$$

Hence

$$(\alpha + 1)(\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1) = 0$$

But  $\alpha \neq -1$ , since  $\alpha = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$  and  $\sin \frac{3\pi}{5} \neq 0$ .

Therefore

$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$$

(d) Since  $\alpha \neq 0$ , divide the result from part (c) by  $\alpha^2$ :

$$\begin{aligned}\frac{\alpha^4}{\alpha^2} - \frac{\alpha^3}{\alpha^2} + \frac{\alpha^2}{\alpha^2} - \frac{\alpha}{\alpha^2} + \frac{1}{\alpha^2} &= 0 \\ \alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} &= 0\end{aligned}$$

Let

$$t = \alpha + \frac{1}{\alpha}$$

Then

$$\begin{aligned}t^2 &= \left(\alpha + \frac{1}{\alpha}\right)^2 \\ &= \alpha^2 + 2 + \frac{1}{\alpha^2}\end{aligned}$$

so

$$\alpha^2 + \frac{1}{\alpha^2} = t^2 - 2$$

Also,

$$\alpha + \frac{1}{\alpha} = t$$

Substituting into

$$\alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} = 0$$

gives

$$\begin{aligned}\left(\alpha^2 + \frac{1}{\alpha^2}\right) - \left(\alpha + \frac{1}{\alpha}\right) + 1 &= 0 \\ (t^2 - 2) - t + 1 &= 0 \\ t^2 - t - 1 &= 0\end{aligned}$$

Therefore,

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$$

(e) From part (d), let

$$t = \alpha + \frac{1}{\alpha}$$

so that

$$t^2 - t - 1 = 0$$

Solving this quadratic,

$$t = \frac{1 \pm \sqrt{5}}{2}$$

Now

$$\alpha = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$$

and since  $|\alpha| = 1$ ,

$$\frac{1}{\alpha} = \cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5}$$

Hence

$$\begin{aligned}\alpha + \frac{1}{\alpha} &= \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}\right) + \left(\cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5}\right) \\ &= 2 \cos \frac{3\pi}{5}\end{aligned}$$

So

$$2 \cos \frac{3\pi}{5} = \frac{1 \pm \sqrt{5}}{2}$$

Therefore

$$\cos \frac{3\pi}{5} = \frac{1 \pm \sqrt{5}}{4}$$

Since  $\frac{3\pi}{5} = 108^\circ$  is in quadrant II,  $\cos \frac{3\pi}{5} < 0$ , so we take the negative value:

$$\cos \frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4} = \frac{1}{4} - \frac{1}{4}\sqrt{5}$$

So  $\cos \frac{3\pi}{5}$  is in the form  $a + b\sqrt{c}$  with  $a = \frac{1}{4}$ ,  $b = -\frac{1}{4}$  and  $c = 5$ .

9. The points representing the complex numbers  $z_1 = 6 + 11i$  and  $z_2 = -8 - 3i$  are opposite vertices of a regular octagon,  $K$ , in the complex plane. The centre of  $K$  represents the complex number  $\alpha$ .

(a) Show that  $\alpha = -1 + 4i$ .

[2]

Given that

$$\beta = \frac{1-i}{14}$$

(b) Show that

$$\beta(z_1 - \alpha) = 1$$

[2]

The vertices of  $K$  are given by the roots of the equation

$$(\beta(z - \alpha))^8 = 1$$

(c) (i) Write down the roots of the equation  $w^8 = 1$  in the form  $re^{i\theta}$ .

[1]

(ii) Hence, or otherwise, determine the positions of the other six vertices of  $K$ , giving your answers as complex numbers in Cartesian form.

[4]

### Solution

(a) Since  $z_1$  and  $z_2$  are opposite vertices of the regular octagon, the centre is the midpoint of the line joining them.

$$\alpha = \frac{z_1 + z_2}{2} = \frac{(6 + 11i) + (-8 - 3i)}{2}$$

$$\alpha = \frac{-2 + 8i}{2} = -1 + 4i$$

So  $\alpha = -1 + 4i$ .

(b) First find  $z_1 - \alpha$ :

$$z_1 - \alpha = (6 + 11i) - (-1 + 4i) = 7 + 7i = 7(1 + i)$$

Now multiply by  $\beta$ :

$$\beta(z_1 - \alpha) = \frac{1-i}{14} \cdot 7(1+i) = \frac{(1-i)(1+i)}{2}$$

Using  $(1-i)(1+i) = 1 - i^2 = 2$ ,

$$\beta(z_1 - \alpha) = \frac{2}{2} = 1$$

Hence  $\beta(z_1 - \alpha) = 1$ .

(c) (i) The roots of  $w^8 = 1$  are the eighth roots of unity:

$$w = e^{ik\pi/4}, \quad k = 0, 1, 2, 3, 4, 5, 6, 7$$

So in the form  $re^{i\theta}$ , we have  $r = 1$  and  $\theta = \frac{k\pi}{4}$ .

(ii) From

$$(\beta(z - \alpha))^8 = 1$$

let

$$w = \beta(z - \alpha)$$

Then  $w^8 = 1$ , so

$$z - \alpha = \frac{w}{\beta} \quad \Rightarrow \quad z = \alpha + \frac{w}{\beta}$$

Now

$$\frac{1}{\beta} = \frac{14}{1-i} = \frac{14(1+i)}{(1-i)(1+i)} = \frac{14(1+i)}{2} = 7(1+i)$$

Hence

$$z = \alpha + 7(1+i)w$$

Using  $\alpha = -1 + 4i$ ,

$$z = -1 + 4i + 7(1+i)w$$

Now substitute the eight roots of  $w^8 = 1$ .

For  $w = 1$ ,

$$z = -1 + 4i + 7(1+i) = 6 + 11i$$

which is  $z_1$ .

For  $w = -1$ ,

$$z = -1 + 4i - 7(1+i) = -8 - 3i$$

which is  $z_2$ .

So the other six vertices come from the remaining six values of  $w$ .

For  $w = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$ ,

$$z = -1 + 4i + 7(1+i)\frac{1+i}{\sqrt{2}} = -1 + 4i + 7\sqrt{2}i$$

so

$$z = -1 + (4 + 7\sqrt{2})i$$

For  $w = e^{i\pi/2} = i$ ,

$$z = -1 + 4i + 7(1+i)i = -1 + 4i + 7(-1+i)$$

so

$$z = -8 + 11i$$

For  $w = e^{3i\pi/4} = \frac{-1+i}{\sqrt{2}}$ ,

$$z = -1 + 4i + 7(1+i)\frac{-1+i}{\sqrt{2}} = -1 + 4i - 7\sqrt{2}i$$

so

$$z = (-1 - 7\sqrt{2}) + 4i$$

For  $w = e^{5i\pi/4} = \frac{-1-i}{\sqrt{2}}$ ,

$$z = -1 + 4i + 7(1+i)\frac{-1-i}{\sqrt{2}} = -1 + 4i - 7\sqrt{2}i$$

so

$$z = -1 + (4 - 7\sqrt{2})i$$

For  $w = e^{3i\pi/2} = -i$ ,

$$z = -1 + 4i + 7(1+i)(-i) = -1 + 4i + 7(1-i)$$

so

$$z = 6 - 3i$$

For  $w = e^{7i\pi/4} = \frac{1-i}{\sqrt{2}}$ ,

$$z = -1 + 4i + 7(1+i)\frac{1-i}{\sqrt{2}} = -1 + 4i + 7\sqrt{2}$$

so

$$z = (-1 + 7\sqrt{2}) + 4i$$

Therefore the other six vertices are

$$\begin{aligned} &-1 + (4 + 7\sqrt{2})i, \quad -8 + 11i, \quad (-1 - 7\sqrt{2}) + 4i, \\ &-1 + (4 - 7\sqrt{2})i, \quad 6 - 3i, \quad (-1 + 7\sqrt{2}) + 4i \end{aligned}$$

10. In an Argand diagram, the points  $A$  and  $B$  are represented by the complex numbers  $1 + 4i$  and  $7 + 2i$  respectively. The circle  $C$  passes through  $A$  and  $B$ , and its centre lies on the real axis.

(a) Find the equation of  $C$ , giving your answer in the form

$$|z - a| = b \quad a \in \mathbb{C}, b \in \mathbb{R} \quad [4]$$

The circle  $D$ , with equation  $|z - (1 + i)| = 5$ , intersects  $C$  at the points representing the complex numbers  $z_1$  and  $z_2$ .

(b) Find the complex numbers  $z_1$  and  $z_2$ . [5]

### Solution

(a) Let the centre of circle  $C$  be the point represented by the real number  $h$ , since the centre lies on the real axis.

So the centre is  $(h, 0)$  on the Argand diagram.

Since  $A(1, 4)$  and  $B(7, 2)$  both lie on  $C$ , their distances from the centre are equal:

$$\begin{aligned}(h - 1)^2 + 4^2 &= (h - 7)^2 + 2^2 \\ (h - 1)^2 + 16 &= (h - 7)^2 + 4\end{aligned}$$

Expanding:

$$\begin{aligned}h^2 - 2h + 1 + 16 &= h^2 - 14h + 49 + 4 \\ -2h + 17 &= -14h + 53 \\ 12h &= 36 \\ h &= 3\end{aligned}$$

So the centre is the complex number 3.

Now find the radius using point  $A$ :

$$\begin{aligned}r^2 &= (3 - 1)^2 + 4^2 \\ &= 2^2 + 4^2 \\ &= 4 + 16 \\ &= 20\end{aligned}$$

Hence

$$r = \sqrt{20} = 2\sqrt{5}$$

Therefore the equation of  $C$  is

$$|z - 3| = 2\sqrt{5}$$

(b) Let

$$z = x + iy$$

Since  $z$  lies on  $C$ , we have

$$|z - 3| = 2\sqrt{5}$$

so

$$(x - 3)^2 + y^2 = 20$$

Since  $z$  also lies on  $D$ , whose equation is  $|z - (1 + i)| = 5$ , we have

$$(x - 1)^2 + (y - 1)^2 = 25$$

So the points of intersection satisfy the simultaneous equations

$$\begin{aligned}(x-3)^2 + y^2 &= 20 \\ (x-1)^2 + (y-1)^2 &= 25\end{aligned}$$

Expand the second equation:

$$\begin{aligned}x^2 - 2x + 1 + y^2 - 2y + 1 &= 25 \\ x^2 + y^2 - 2x - 2y - 23 &= 0\end{aligned}$$

Expand the first equation:

$$\begin{aligned}x^2 - 6x + 9 + y^2 &= 20 \\ x^2 + y^2 - 6x - 11 &= 0\end{aligned}$$

Subtract the first from the second:

$$\begin{aligned}(x^2 + y^2 - 2x - 2y - 23) - (x^2 + y^2 - 6x - 11) &= 0 \\ -2x - 2y - 23 + 6x + 11 &= 0 \\ 4x - 2y - 12 &= 0 \\ 2x - y - 6 &= 0\end{aligned}$$

Hence

$$y = 2x - 6$$

Substitute into  $(x-3)^2 + y^2 = 20$ :

$$\begin{aligned}(x-3)^2 + (2x-6)^2 &= 20 \\ (x-3)^2 + 4(x-3)^2 &= 20 \\ 5(x-3)^2 &= 20 \\ (x-3)^2 &= 4\end{aligned}$$

So

$$x - 3 = \pm 2$$

giving

$$x = 1 \quad \text{or} \quad x = 5$$

Now use  $y = 2x - 6$ .

If  $x = 1$ ,

$$y = 2(1) - 6 = -4$$

If  $x = 5$ ,

$$y = 2(5) - 6 = 4$$

Therefore the two intersection points are represented by

$$z_1 = 1 - 4i, \quad z_2 = 5 + 4i$$

11. The cubic equation

$$z^3 + pz^2 + qz + 24 = 0$$

where  $p$  and  $q$  are real, has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

It is given that  $\alpha = -2 + 2i$ .

(a) (i) Write down another root,  $\beta$ , of the equation. [1]

(ii) Find the third root,  $\gamma$ . [3]

(iii) Find the values of  $p$  and  $q$ . [3]

(b) (i) Express  $\alpha$  in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ . [2]

(ii) Hence show that

$$(-2 + 2i)^n = (2\sqrt{2})^n \left( \cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right) \quad [2]$$

(iii) Hence show that

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos \frac{3n\pi}{4} + (-3)^n$$

where  $n$  is an integer. [3]

### Solution

(a) (i) Since the coefficients of the cubic are real, complex roots occur in conjugate pairs.

So the other complex root is

$$\beta = -2 - 2i$$

(ii) For the monic cubic

$$z^3 + pz^2 + qz + 24 = 0$$

the product of the roots is

$$\alpha\beta\gamma = -24$$

First find  $\alpha\beta$ :

$$\begin{aligned} \alpha\beta &= (-2 + 2i)(-2 - 2i) \\ &= (-2)^2 - (2i)^2 \\ &= 4 - (-4) \\ &= 8 \end{aligned}$$

So

$$8\gamma = -24$$

and hence

$$\gamma = -3$$

(iii) Using the relationships between roots and coefficients:

$$\alpha + \beta + \gamma = -p$$

Now

$$\begin{aligned} \alpha + \beta + \gamma &= (-2 + 2i) + (-2 - 2i) + (-3) \\ &= -7 \end{aligned}$$

so

$$-p = -7$$

giving

$$p = 7$$

Also

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

We already know that  $\alpha\beta = 8$ , and

$$\begin{aligned}\beta\gamma + \gamma\alpha &= \gamma(\alpha + \beta) \\ &= (-3)((-2 + 2i) + (-2 - 2i)) \\ &= (-3)(-4) \\ &= 12\end{aligned}$$

Therefore

$$q = 8 + 12 = 20$$

So the values are

$$p = 7, \quad q = 20$$

(b) (i) For

$$\alpha = -2 + 2i$$

its modulus is

$$\begin{aligned}r &= |\alpha| \\ &= \sqrt{(-2)^2 + 2^2} \\ &= \sqrt{8} \\ &= 2\sqrt{2}\end{aligned}$$

The point  $(-2, 2)$  lies in the second quadrant, so the principal argument is

$$\theta = \frac{3\pi}{4}$$

Hence

$$\alpha = 2\sqrt{2} e^{i3\pi/4}$$

(ii) From part (i),

$$\alpha = 2\sqrt{2} e^{i3\pi/4}$$

so

$$\begin{aligned}\alpha^n &= \left(2\sqrt{2} e^{i3\pi/4}\right)^n \\ &= (2\sqrt{2})^n e^{i3n\pi/4}\end{aligned}$$

Using  $e^{i\theta} = \cos \theta + i \sin \theta$ ,

$$(-2 + 2i)^n = (2\sqrt{2})^n \left( \cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right)$$

(iii) Since

$$\beta = -2 - 2i = 2\sqrt{2} e^{-i3\pi/4}$$

we have

$$\begin{aligned}\beta^n &= \left(2\sqrt{2} e^{-i3\pi/4}\right)^n \\ &= (2\sqrt{2})^n \left( \cos \left( -\frac{3n\pi}{4} \right) + i \sin \left( -\frac{3n\pi}{4} \right) \right) \\ &= (2\sqrt{2})^n \left( \cos \frac{3n\pi}{4} - i \sin \frac{3n\pi}{4} \right)\end{aligned}$$

Therefore

$$\begin{aligned}\alpha^n + \beta^n &= (2\sqrt{2})^n \left( \cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right) + (2\sqrt{2})^n \left( \cos \frac{3n\pi}{4} - i \sin \frac{3n\pi}{4} \right) \\ &= 2(2\sqrt{2})^n \cos \frac{3n\pi}{4}\end{aligned}$$

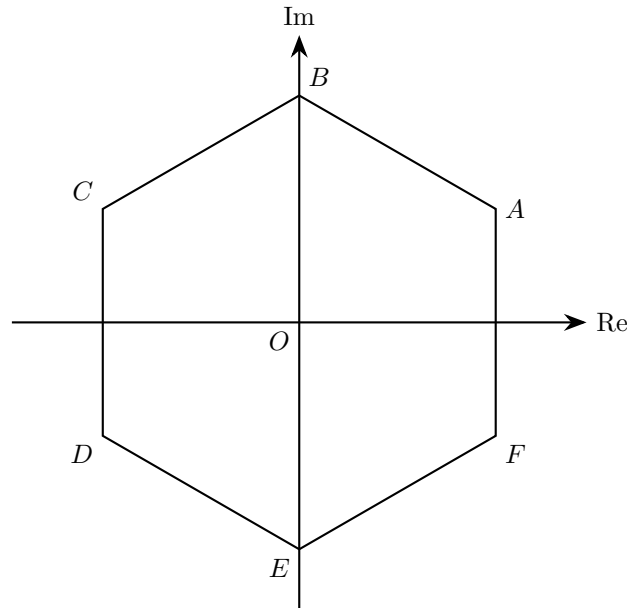
Also, from part (a),  $\gamma = -3$ , so

$$\gamma^n = (-3)^n$$

Hence

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos \frac{3n\pi}{4} + (-3)^n$$

as required.



12. In the diagram, the points  $A, B, C, D, E$  and  $F$  represent the complex sixth roots of  $-729$  on an Argand diagram. The centroids of triangles  $OAB, OBC, OCD, ODE, OEF$  and  $OFA$  are  $G, H, I, J, K$  and  $L$  respectively.

- (a) Write down, in exponential form, the complex numbers represented by the points  $A, B, C, D, E$  and  $F$ . [2]
- (b) When these complex numbers are multiplied by the complex number  $w$ , the resulting complex numbers are represented by the points  $G, H, I, J, K$  and  $L$ . Find  $w$  in exponential form. [4]
- (c) You are given that  $G, H, I, J, K$  and  $L$  represent roots of the equation  $z^6 = p$ . Find  $p$ . [2]

### Solution

- (a) Since

$$-729 = 729e^{i\pi} = 3^6 e^{i(\pi+2k\pi)}$$

the sixth roots are

$$z = 3e^{i(\frac{\pi+2k\pi}{6})} = 3e^{i(\frac{\pi}{6} + \frac{k\pi}{3})}, \quad k = 0, 1, 2, 3, 4, 5$$

Taking these in order round the Argand diagram gives

$$\begin{aligned} A &= 3e^{i\pi/6}, & B &= 3e^{i\pi/2}, & C &= 3e^{i5\pi/6}, \\ D &= 3e^{i7\pi/6}, & E &= 3e^{i3\pi/2}, & F &= 3e^{i11\pi/6} \end{aligned}$$

- (b) The centroid of triangle  $OAB$  is

$$G = \frac{O + A + B}{3} = \frac{A + B}{3}$$

If multiplication by  $w$  maps  $A$  to  $G$ , then

$$Aw = G = \frac{A + B}{3}$$

so

$$w = \frac{A + B}{3A}$$

Now  $B$  is the next sixth root after  $A$ , so

$$B = Ae^{i\pi/3}$$

Hence

$$\begin{aligned} w &= \frac{A + Ae^{i\pi/3}}{3A} \\ &= \frac{1 + e^{i\pi/3}}{3} \end{aligned}$$

Now

$$e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

so

$$1 + e^{i\pi/3} = \frac{3}{2} + \frac{\sqrt{3}}{2}i$$

Its modulus is

$$\sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{3}$$

and its argument is

$$\tan^{-1}\left(\frac{\sqrt{3}/2}{3/2}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

So

$$1 + e^{i\pi/3} = \sqrt{3}e^{i\pi/6}$$

Therefore

$$w = \frac{\sqrt{3}}{3}e^{i\pi/6} = \frac{1}{\sqrt{3}}e^{i\pi/6}$$

Also, for example,

$$H = \frac{B + C}{3} = \frac{B + Be^{i\pi/3}}{3} = Bw$$

and similarly for the other consecutive pairs, so this same  $w$  maps  $A, B, C, D, E, F$  to  $G, H, I, J, K, L$ .

Hence

$$w = \frac{1}{\sqrt{3}}e^{i\pi/6}$$

(c) Let  $r$  be any sixth root of  $-729$ . Then the corresponding point among  $G, H, I, J, K, L$  is

$$z = wr$$

So

$$z^6 = w^6 r^6 = w^6 (-729)$$

Now

$$\begin{aligned} w^6 &= \left(\frac{1}{\sqrt{3}}e^{i\pi/6}\right)^6 \\ &= \left(\frac{1}{\sqrt{3}}\right)^6 e^{i\pi} \\ &= \frac{1}{27}(-1) \\ &= -\frac{1}{27} \end{aligned}$$

Hence

$$p = w^6(-729) = \left(-\frac{1}{27}\right)(-729) = 27$$

13. (i) Write down, in any form, all the complex roots of the equation

$$w^{11} = -1 \quad [2]$$

- (ii) Explain why the equation

$$(z+1)^{11} = -(z-1)^{11}$$

has exactly 10 non-real roots and show that they may be expressed in the form

$$-i \cot\left(\frac{m\pi}{22}\right) \quad [6]$$

where  $m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$ .

- (iii) Show that

$$\left\{-i \cot\left(\frac{m\pi}{22}\right)\right\} \left\{i \cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right) \quad [1]$$

- (iv) Given that the product of the non-zero roots of the equation in part (ii) is 11, find the value of

$$\tan^2\left(\frac{\pi}{22}\right) \tan^2\left(\frac{3\pi}{22}\right) \tan^2\left(\frac{5\pi}{22}\right) \tan^2\left(\frac{7\pi}{22}\right) \tan^2\left(\frac{9\pi}{22}\right) \quad [2]$$

### Solution

- (i) Write  $-1$  in exponential form as

$$-1 = e^{i(2k+1)\pi} \quad (k \in \mathbb{Z})$$

So if  $w^{11} = -1$ , then

$$w = e^{i(2k+1)\pi/11}$$

Taking the 11 distinct values gives

$$w = e^{i(2k+1)\pi/11}, \quad k = 0, 1, \dots, 10$$

So all the complex roots are

$$e^{i\pi/11}, e^{3i\pi/11}, e^{5i\pi/11}, \dots, e^{21i\pi/11}$$

- (ii) First note that  $z = 1$  is not a root of

$$(z+1)^{11} = -(z-1)^{11}$$

because this would give  $2^{11} = 0$ , which is false.

So we may divide by  $(z-1)^{11}$ :

$$\left(\frac{z+1}{z-1}\right)^{11} = -1$$

Let

$$w = \frac{z+1}{z-1}$$

Then

$$w^{11} = -1$$

From part (i), the 11 possible values of  $w$  are

$$w = e^{i\theta}, \quad \theta = \frac{(2k+1)\pi}{11}, \quad k = 0, 1, \dots, 10$$

Now solve for  $z$  in terms of  $w$ :

$$\begin{aligned} w &= \frac{z+1}{z-1} \\ w(z-1) &= z+1 \\ wz - w &= z+1 \\ z(w-1) &= w+1 \\ z &= \frac{w+1}{w-1} \end{aligned}$$

So

$$z = \frac{e^{i\theta} + 1}{e^{i\theta} - 1}$$

Now multiply top and bottom by  $e^{-i\theta/2}$ :

$$\begin{aligned} z &= \frac{e^{i\theta/2} + e^{-i\theta/2}}{e^{i\theta/2} - e^{-i\theta/2}} \\ &= \frac{2 \cos(\theta/2)}{2i \sin(\theta/2)} \\ &= -i \cot\left(\frac{\theta}{2}\right) \end{aligned}$$

Since  $\theta = \frac{(2k+1)\pi}{11}$ , this gives

$$z = -i \cot\left(\frac{(2k+1)\pi}{22}\right) \quad k = 0, 1, \dots, 10$$

So the 11 roots are obtained from

$$m = 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21$$

in

$$z = -i \cot\left(\frac{m\pi}{22}\right)$$

When  $m = 11$ ,

$$z = -i \cot\left(\frac{11\pi}{22}\right) = -i \cot\left(\frac{\pi}{2}\right) = 0$$

so this is the one real root.

To see there are no other real roots, suppose  $z$  is real. Then

$$w = \frac{z+1}{z-1}$$

is also real. But the only real solution of  $w^{11} = -1$  is  $w = -1$ , which gives

$$z = \frac{-1+1}{-1-1} = 0$$

Hence there is exactly one real root, namely 0, so the remaining  $11 - 1 = 10$  roots are non-real.

Now rewrite the values with  $m = 13, 15, 17, 19, 21$ . If  $r = 1, 3, 5, 7, 9$ , then

$$\begin{aligned} -i \cot\left(\frac{(22-r)\pi}{22}\right) &= -i \cot\left(\pi - \frac{r\pi}{22}\right) \\ &= -i \left(-\cot\left(\frac{r\pi}{22}\right)\right) \\ &= i \cot\left(\frac{r\pi}{22}\right) \\ &= -i \cot\left(-\frac{r\pi}{22}\right) \end{aligned}$$

Therefore the 10 non-real roots may be written as

$$-i \cot\left(\frac{m\pi}{22}\right), \quad m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$$

(iii) Let

$$\alpha = \frac{m\pi}{22}$$

Then

$$\begin{aligned}(-i \cot \alpha)(i \cot \alpha) &= (-i \cdot i) \cot^2 \alpha \\ &= -i^2 \cot^2 \alpha \\ &= \cot^2 \alpha\end{aligned}$$

So

$$\left\{-i \cot \left(\frac{m\pi}{22}\right)\right\} \left\{i \cot \left(\frac{m\pi}{22}\right)\right\} = \cot^2 \left(\frac{m\pi}{22}\right)$$

(iv) The non-zero roots occur in pairs

$$-i \cot \left(\frac{m\pi}{22}\right) \quad \text{and} \quad -i \cot \left(-\frac{m\pi}{22}\right) = i \cot \left(\frac{m\pi}{22}\right)$$

for

$$m = 1, 3, 5, 7, 9$$

Given that the product of the non-zero roots is 11,

$$\begin{aligned}11 &= \prod_{m=1,3,5,7,9} \left\{-i \cot \left(\frac{m\pi}{22}\right)\right\} \left\{i \cot \left(\frac{m\pi}{22}\right)\right\} \\ &= \prod_{m=1,3,5,7,9} \cot^2 \left(\frac{m\pi}{22}\right)\end{aligned}$$

Therefore

$$\cot^2 \left(\frac{\pi}{22}\right) \cot^2 \left(\frac{3\pi}{22}\right) \cot^2 \left(\frac{5\pi}{22}\right) \cot^2 \left(\frac{7\pi}{22}\right) \cot^2 \left(\frac{9\pi}{22}\right) = 11$$

Taking reciprocals,

$$\tan^2 \left(\frac{\pi}{22}\right) \tan^2 \left(\frac{3\pi}{22}\right) \tan^2 \left(\frac{5\pi}{22}\right) \tan^2 \left(\frac{7\pi}{22}\right) \tan^2 \left(\frac{9\pi}{22}\right) = \frac{1}{11}$$